

INSPECT

INSPECT Syntax - Format 1

INSPECT SourceStr\$i TALLYING

Counter# i FOR { CHARACTERS [{ BEFORE } INITIAL Delim\$il] ...
 { ALL } { Compare\$il [{ BEFORE } INITIAL Delim\$il] ... } ... } ...
 { LEADING }

INSPECT FullName TALLYING UnstrPtr
FOR LEADING SPACES.

INSPECT SourceLine TALLYING ECount
FOR ALL "e" AFTER INITIAL "start"
BEFORE INITIAL "end".



INSPECT Example 1

```
END-TEXTFILE
  AT END SET EndOfFile TO TRUE
END-READ
PERFORM UNTIL EndOfFile
  PERFORM VARYING idx FROM 1 BY 1 UNTIL idx > 26
    INSPECT TextLine TALLYING LetterCount(idx)
      FOR ALL Letter(idx)
  END-PERFORM
  READ TextFile
  AT END SET EndOfFile TO TRUE
END-READ
END-PERFORM
PERFORM VARYING idx FROM 1 BY 1 UNTIL idx > 26
  DISPLAY "Letter " Letter(idx)
    " occurs " LetterCount(idx) " times"
END-PERFORM
```

How the INSPECT works.

The INSPECT scans the **Source String** from left to right **counting** and/or **replacing** characters under the control of the **TALLYING**, **REPLACING** or **CONVERTING** phrases.

The **behaviour** of the INSPECT is **modified by** using the **LEADING**, **FIRST**, **BEFORE** and **AFTER** phrases.

An **ALL**, **LEADING**, **CHARACTERS**, **FIRST** or **CONVERTING** phrase may only be followed by **one BEFORE** and **one AFTER** phrase.



Modifying Phrases

LEADING

The LEADING phrase causes counting/replacement of all Compare\$il characters from the **first valid one** encountered to the **first invalid one**.

FIRST

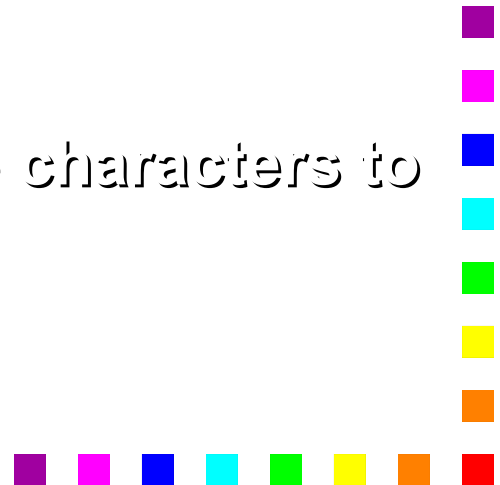
The FIRST phrase causes only the **first valid character** to be replaced.

BEFORE

The BEFORE phrase designates as valid those characters to the **left of the delimiter** associated with it.

AFTER

The AFTER phrase designates as valid those characters to the **right of the delimiter** associated with it.



INSPECT Syntax - Format 2

INSPECT SourceStr\$i REPLACING

$$\left\{ \begin{array}{l} \text{CHARACTERS BY ReplaceChar\$il} \left[\left\{ \begin{array}{c} \text{BEFORE} \\ \text{AFTER} \end{array} \right\} \text{INITIAL Delim\$il} \right] \dots \\ \left\{ \begin{array}{c} \text{ALL} \\ \text{LEADING} \\ \text{FIRST} \end{array} \right\} \left\{ \text{Compare\$il BY Replace\$il} \left[\left\{ \begin{array}{c} \text{BEFORE} \\ \text{AFTER} \end{array} \right\} \text{INITIAL Delim\$il} \right] \dots \right\} \dots \end{array} \right\} \dots$$

```
PERFORM VARYING idx FROM 1 BY 1 UNTIL idx > 10
  INSPECT TextLine
    REPLACING SwearWord(idx) BY "*#@!"
END-PERFORM
```



INSPECT Example 2

INSPECT StringData READING ALL F'S BEFORE INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 2

INSPECT StringData READING ALL F'S BEFORE INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	G	G	G	G	G	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 3

INSPECT StringData REPLACING LEADING "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Z".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 3

INSPECT StringData REPLACING LEADING "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Z".

StringData

F	F	F	F	A	G	G	G	G	G	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 4

INSPECT StringData READING ALL F'S BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Z".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 4

INSPECT StringData READING ALL F'S BEFORE INITIAL "A" BEFORE INITIAL "Z".

StringData

F	F	F	F	A	G	G	G	G	G	Q	G	G	G	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 5

INSPECT StringData REMOVING FIRST "F" BEFORE INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 5

INSPECT StringData REMAINING FIRST "F" BEFORE INITIAL "A" BEFORE INITIAL "G".

StringData

F	F	F	F	A	G	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 6

INSPECT StringData WITH ACEND
ALL "FFFF" BY "FROG"
AFTER INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 6

INSPECT StringData WITH ACEND
ALL "FFFF" BY "FROG"
AFTER INITIAL "A" BEFORE INITIAL "Q".

StringData

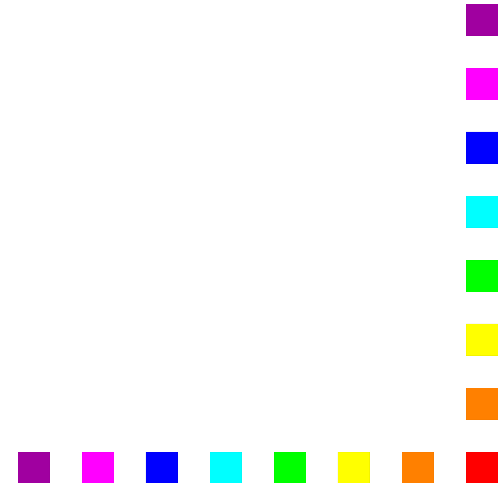
F	F	F	F	A	F	R	O	G	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Syntax - Format 3

INSPECT SourceStr\$i TALLYING

Counter# i FOR $\left\{ \begin{array}{l} \underline{\text{CHARACTERS}} \left[\begin{array}{l} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right] \text{INITIAL Delim$il} \dots \\ \left\{ \begin{array}{l} \underline{\text{ALL}} \\ \underline{\text{LEADING}} \end{array} \right\} \left\{ \text{De lim $il} \left[\begin{array}{l} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right] \text{INITIAL Delim$il} \dots \right\} \dots \end{array} \right\} \dots$

REPLACING $\left\{ \begin{array}{l} \underline{\text{CHARACTERS}} \text{ BY ReplaceChar$il} \left[\begin{array}{l} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right] \text{INITIAL Delim$il} \dots \\ \left\{ \begin{array}{l} \underline{\text{ALL}} \\ \underline{\text{LEADING}} \\ \underline{\text{FIRST}} \end{array} \right\} \left\{ \text{Compare$il BY Replace$il} \left[\begin{array}{l} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right] \text{INITIAL Delim$il} \dots \right\} \dots \end{array} \right\} \dots$



INSPECT Syntax - Format 4

INSPECT SourceStr\$*i* CONVERTING

Compare\$*il* TO Convert\$*il*

[{BEFORE } INITIAL Delim\$*il*] ...

INSPECT TextLine CONVERTING

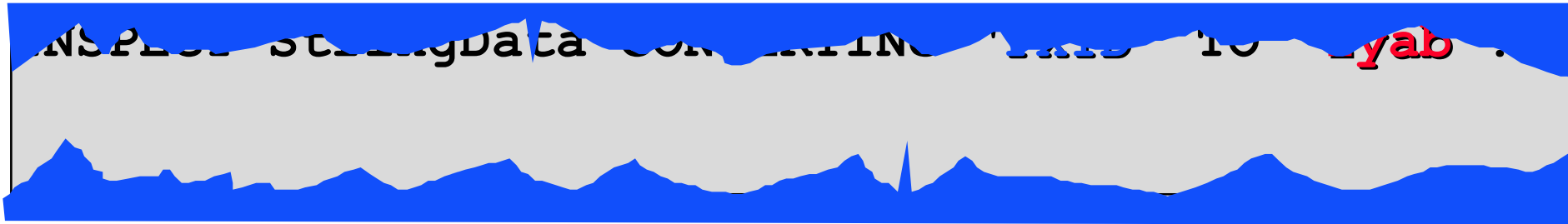
"0123456789" TO "5298317046"

AFTER INITIAL "codeon"

BEFORE INITIAL "codeoff".



INSPECT Example 7



StringData

P	X	P	F	P	D	P	T	P	P	F	X	T	D	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 7



StringData

P	y	P	z	P	b	P	a	P	P	z	y	a	b	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 7

INSPECT StringData REPLACING "FX" BY "zyab".

StringData

P	X	P	F	P	D	P	T	P	P	F	X	T	D	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 7

INSPECT StringData REPLACING "FAD" BY "zyab".

StringData

P	X	P	F	P	D	P	T	P	P	z	y	a	b	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 8



StringData

P	x	P	f	P	d	P	T	P	P	f	x	T	d	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 8



StringData

P	y	P	s	P	z	P	T	P	P	s	y	T	z	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 9

INSPECT CustAddress

CONVERTING "abcdefghijklmnopqrstuvwxyz"
TO "ABCDEFGHIJKLMNOPQRSTUVWXYZ".

01 AlphaChars.

02 AlphaLower PIC X(26) VALUE
"abcdefghijklmnopqrstuvwxyz".

02 AlphaUpper PIC X(26) VALUE
"ABCDEFGHIJKLMNOPQRSTUVWXYZ".

INSPECT CustAddress

CONVERTING AlphaLower TO AlphaUpper.

INSPECT CustAddress

CONVERTING AlphaUpper TO AlphaLower.

INSPECT Example 10

```
01 RussianPay PIC $$$,###.99.
```

```
MOVE 12345.67 TO RussianPay.  
INSPECT RussianPay REPLACING ALL "$" BY "R".  
DISPLAY RussianPay " roubles".
```

RussianPay



INSPECT Example 10

```
1 RussianPay PIC $$$,###.99.
```

```
MOVE 12345.67 TO RussianPay.
```

```
INSPECT RussianPay REPLACING ALL "$" BY "R".
```

```
DISPLAY RussianPay " roubles".
```

RussianPay

\$12,345.67

INSPECT Example 10

```
1 12345.67 RussianPay PIC $$$,###.##.
```

```
MOVE 12345.67 TO RussianPay.
```

```
INSPECT RussianPay REPLACING ALL "$" BY "R".
```

```
DISPLAY RussianPay " roubles".
```

RussianPay

R12,345.67

INSPECT Example 10

```
1 12345.67 RussianPay PIC $$$,$$.99.
```

```
MOVE 12345.67 TO RussianPay.  
INSPECT RussianPay REPLACING ALL "$" BY "R".  
DISPLAY RussianPay " roubles".
```

RussianPay

R12,345.67

R12,345.67 roubles